

JOVE-Core Synthetic Pipeline Dry Run

Engineering Validation Only

August 2026

Warning: This report contains generated labels and simulated system outputs. It includes no real evidence, participants, independent ratings, or empirical result.

1 Purpose

Validate grouped data generation, B0–V4 scoring, coverage/risk accounting, clustered bootstrap, figure generation, and manuscript assembly before live data collection.

2 Synthetic configuration

Thirty simulated base cases, eight variants per base case, six systems, and 1,440 generated outputs. Apparent effect sizes were intentionally encoded by the simulator and have no scientific meaning.

3 Pipeline output

4 Primary bootstrap test

The dry run produced a synthetic FAR difference and confidence interval. This verifies code execution only; statistical significance is predetermined by the generator assumptions.

5 Next gate

Replace all synthetic inputs with consented live cases, independent ratings, frozen model outputs, and sealed confirmatory labels. The submission gate rejects this report's provenance.

Table 1: Confirmatory system results. Values are generated only from the frozen scoring pipeline.

System	FAR ↓	FRR ↓	Coverage ↑	Risk ↓
B0_generic_judge	0.626	0.120	1.000	1.663
B1_policy_judge	0.529	0.081	1.000	1.192
V1_structured	0.544	0.095	1.000	1.221
V2_forensic	0.394	0.101	1.000	0.654
V3_ensemble	0.315	0.060	1.000	0.521
V4_selective	0.163	0.075	0.813	0.262